# Quantum Random Walks

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# A crude introduction to Qubits

- Imagine we have two coins and we wish to construct a number between 0 and 3: Toss each coin, and using the convention that a head maps to 0 and a tail maps to 1, we get the following mapping:

$$\{HH, HT, TH, TT\} \leftrightarrow \{0, 1, 2, 3\}$$

- Now, how qubits work is that the coins have not landed on either side, and are still spinning. This means that neither of the coins has resulted in a heads or a tails, but is "contributing to both of them" [read as: superposition]. This allows both coins to "simultaneously represent" all 4 numbers $\{0, 1, 2, 3\}$. However, as soon as either of these coins is "observed", they will collapse into a heads or a tails.
- Qubits allow us to operate on these superpositions, i.e we can operate on all the $2^n$ numbers simultaneously.

# More about Superpositions

- The state of each qubit can be given by

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

  where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.
- Upon observation, the probability of obtaining $|0\rangle$ is $|\alpha|^2$and that of observing $|1\rangle$ is $|\beta|^2$
- There is absolutely no way to tell the values of $\alpha$ and $\beta$!
  - **Idea**: Recreate the superposition and measure multiple times to approximate the probability of obtaining $|0\rangle$ and $|1\rangle$.
  - **Problem**: No Cloning Theorem: if we have two states $|\psi\rangle, |\phi\rangle$, a pure state $|s\rangle$, and a unitary operator $U$ such that

$$U\,|\phi\rangle\,|s\rangle = |\phi\rangle\,|\phi\rangle$$

    then $|\phi\rangle$ *and* $|\psi\rangle$ are either equal or orthogonal to each other.

# Multiple qubits

- We can choose a computational basis to work in. For our convenience, we choose the standard basis, i.e

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

  Thus, any superposition $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ can be written as $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

- If we have two qubits, we shall have 4 possible states, i.e we can take a Hadamard product to obtain the 4 states: if $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ and $|\phi\rangle = \gamma |0\rangle + \delta |1\rangle$, then

$$\begin{aligned} |\psi\rangle \otimes |\phi\rangle &= (\alpha |0\rangle + \beta |1\rangle) \otimes (\gamma |0\rangle + \delta |1\rangle) \\ &= \alpha\gamma |0\rangle \otimes |0\rangle + \alpha\delta |0\rangle \otimes |1\rangle + \beta\gamma |1\rangle \otimes |0\rangle + \beta\delta |1\rangle \otimes |1\rangle \end{aligned}$$

# Multiple qubits continued

- We can write the following:

$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Similarly, we can get

$$|0\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, |1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

# Multiple qubits continued

- Thus, our new state would be

$$
\begin{aligned}
|\psi\rangle \otimes |\phi\rangle &= (\alpha |0\rangle + \beta |1\rangle) \otimes (\gamma |0\rangle + \delta |1\rangle) \\
&= \alpha\gamma |0\rangle \otimes |0\rangle + \alpha\delta |0\rangle \otimes |1\rangle + \beta\gamma |1\rangle \otimes |0\rangle + \beta\delta |1\rangle \otimes |1\rangle \\
&= \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}
\end{aligned}
$$

- In general, having $n$ qubits allows us to work in a $2^n$ dimensional Hilbert space, and the primary operation that allows us to do that is the Kronecker or the Tensor product.
- A useful property:

$$(A \otimes B)(C \otimes D) = AC \otimes BD$$

# Changing the computational basis

- We can also change the computational basis. Take, for example the Hadamard matrix given by

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- We can check $H|0\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $H|1\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$
- We denote

$$|+\rangle = H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- Note that $|+\rangle$ and $|-\rangle$ are orthogonal basis states too!

# Notation

- $|0\rangle$ is called as a "ket" vector and is used to represent a column vector
- $\langle 0|$ is called as a "bra" vector and is used to represent a row vector.
- Since we work in the complex field,

$$\langle 0| = |0\rangle^{\dagger}$$

  where $\dagger$ represents the complex conjugate transpose.
- Since we are operating in a Hilbert field, the inner product exists and thus,

$$\langle 0|\,|0\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

  For convenience, this is written as $\langle 0|0\rangle$
- Also for the sake of convenience,

$$|\phi\rangle \otimes |\psi\rangle = |\phi\psi\rangle$$

# Projections

- We first do a corresponding linear algebra problem: given $\vec{a}$ that has to be projected along $\vec{b}$, what is the projection matrix?
- $\vec{a}$ can be decomposed into a component along $\vec{b}$, say $\beta\vec{b}$ and a component perpendicular to $\vec{b}$, say $\vec{c}$. Then

$$\vec{a} = \beta\vec{b} + \vec{c} \implies \frac{\vec{b}^T\vec{a}}{\vec{b}^T\vec{b}} = \beta$$

$$\implies proj_b\vec{a} = \left(\frac{\vec{b}^T\vec{a}}{\vec{b}^T\vec{b}}\right)\vec{b} = \left(\frac{\vec{b}\vec{b}^T}{\vec{b}^T\vec{b}}\right)\vec{a}$$

- Notation equivalent:

$$\frac{|b\rangle\langle b|}{\langle b|b\rangle} = |b\rangle\langle b|$$

since $\langle b|b\rangle = 1$

# Measurements

- Suppose we wish to measure for basis state $|\phi\rangle$ in the overall superposition $|\psi\rangle$. Then, the corresponding projection operator is given $P_\phi = |\phi\rangle\langle\phi|$
- The probability of obtaining $|\phi\rangle$ is given by

$$\langle\psi| P_\phi |\psi\rangle$$

using the idea of quadratic forms! What this does is projects the state onto $|\phi\rangle$ and then takes an inner product with itself, resulting in the probability!

- The state after measurement is given by

$$|\bar{\psi}\rangle = \frac{P_\phi |\psi\rangle}{\sqrt{\langle\psi| P_\phi |\psi\rangle}}$$

by collapsing the state onto $|\phi\rangle$ and re-normalizing the state.

# Grover's Search Algorithm

- One of the revolutionary algorithms in the field of quantum computing!
- Imagine having a binary string of length $N$ with $t$ ones in it, and we wish to locate the indices at which 1 is present.
- Deterministic and randomized classical algorithms require $\mathcal{O}(N)$ queries.
- Grover's Algorithm can do the same in $\mathcal{O}(\sqrt{N})$ queries.
- Create a superposition over all states:

$$|\psi\rangle = H^{\otimes N}\left|0^N\right\rangle = \frac{1}{\sqrt{N}}\sum_{i=0}^{2^N-1}|i\rangle$$

- Divide these into "Good" and "Bad" superpositions

$$|G\rangle = \frac{1}{\sqrt{t}}\sum_{i:x_i=1}|i\rangle \quad |B\rangle = \frac{1}{\sqrt{N-t}}\sum_{i:x_i=0}|i\rangle$$

# Phase Query

- We can rewrite the superposition as

$$|\psi\rangle = \sqrt{\frac{t}{N}}\,|G\rangle + \sqrt{\frac{N-t}{N}}\,|B\rangle = \sin\theta\,|G\rangle + \cos\theta\,|B\rangle$$

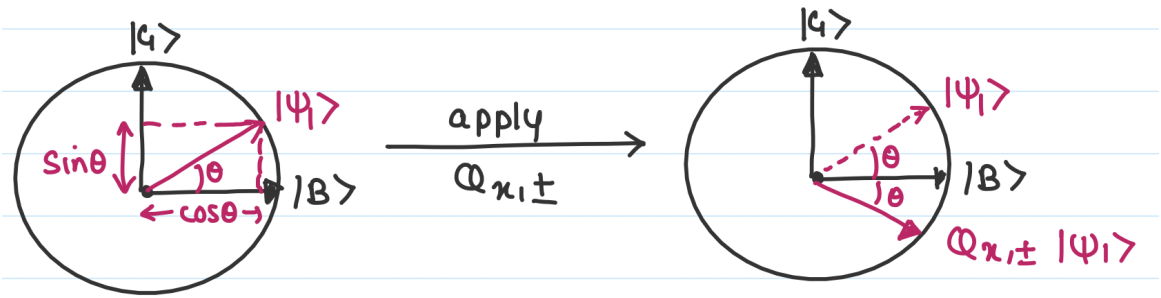
- The phase query acts as follows:

$$Q\,|i\rangle = (-1)^{x_i}\,|i\rangle$$

- Applying the phase query on the superposition results in

$$Q\,|\psi\rangle = -\sin\theta\,|G\rangle + \cos\theta\,|B\rangle$$

- Note that $|G\rangle$ and $|B\rangle$ are orthogonal and the phase query results in a reflection about $|B\rangle$



# Grover's Iterate

- The Grover iterate is given by $Gr = H^{\otimes N} R_0 H^{\otimes N}$ where $R_0$ is a reflection about $\left|0^N\right\rangle$, i.e for all non-zero states $R_0 \left|i\right\rangle = -\left|i\right\rangle$
- A simple result from linear algebra: the reflection matrix about $\vec{b}$ is given by $2P_b - I$ where $P_b = \left|\vec{b}\right\rangle\left\langle\vec{b}\right|$
  - **Proof**: Let $\vec{a} = P_b\vec{a} + \vec{c}$ where $\langle\vec{c}, \vec{a}\rangle = 0$
  - Then, $ref_b\vec{a} = P_b\vec{a} - \vec{c} = 2P_b\vec{a} - \vec{a} = (2P_b - 1)\vec{a}$
- We have

$$\begin{aligned} Gr = H^{\otimes N} R_0 H^{\otimes N} &= H^{\otimes N}(2\left|0^N\right\rangle\left\langle 0^N\right| - I)H^{\otimes N} \\ &= 2H^{\otimes N}(2\left|0^N\right\rangle\left\langle 0^N\right|)H^{\otimes N} - H^{\otimes N} I H^{\otimes N} \\ &= 2\left|\psi\right\rangle\left\langle\psi\right| - I \end{aligned}$$

i.e the Grover's iterate is a reflection about the original state $\left|\psi\right\rangle$

# Grover's Iterate continued

- Thus, applying the Grover's iterate results in:
  $Gr(Q|\psi\rangle) = Gr(-\sin\theta|G\rangle + \cos\theta|B\rangle) = \sin 3\theta|G\rangle + \cos 3\theta|B\rangle$

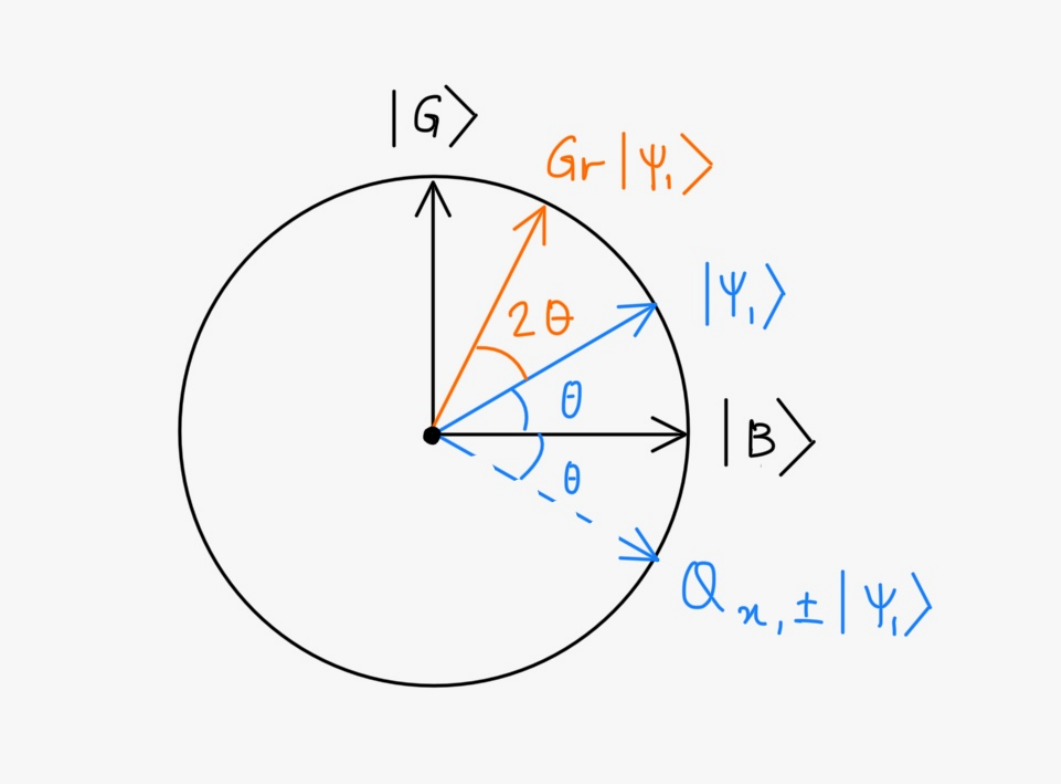



Figure: Applying the Grover's iterate

# The Algorithm

- Apply the phase query to $|\psi_0\rangle = \sin\theta\,|G\rangle + \cos\theta\,|B\rangle$ to get

$$\left|\tilde{\psi_0}\right\rangle = -\sin\theta\,|G\rangle + \cos\theta\,|B\rangle$$

  $\tilde{\psi_0}$ is at an angle of $2\theta$ with $|\psi_0\rangle$
- Apply the Grover's iterate to $\left|\tilde{\psi_0}\right\rangle$ to obtain

$$|\psi_1\rangle = \sin 3\theta\,|G\rangle + \cos 3\theta\,|B\rangle$$

- Apply the phase query to $|\psi_1\rangle$ to get

$$\left|\tilde{\psi_1}\right\rangle = -\sin 3\theta\,|G\rangle + \cos 3\theta\,|B\rangle$$

  $\tilde{\psi_0}$ is at an angle of $4\theta$ with $|\psi_0\rangle$
- Apply the Grover's iterate to $\left|\tilde{\psi_1}\right\rangle$ to obtain

$$|\psi_2\rangle = \sin 5\theta\,|G\rangle + \cos 5\theta\,|B\rangle$$

  and so on...

# Correctness

- Probability of getting a good term is $\sin^2(2k+1)\theta$. This is maximum when the number of iterations $k = \left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$
- Assume $\tilde{k} = \left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.We choose $k$ to be the closest integer to $\tilde{k}$
- The probability of error is given by:

$$\begin{aligned}
\cos^2(2k+1)\theta &= \cos^2((2\tilde{k}+1)\theta + 2(k-\tilde{k})\theta) \\
&= \cos^2\left(\frac{\pi}{2} + \leq \theta\right) \\
&\leq \sin^2\theta \\
&\leq \frac{t}{N}
\end{aligned}$$

# Discrete Quantum Random Walks

- The state space is given by $\mathcal{H} = \mathcal{H}_c \otimes \mathcal{H}_p$ where $\mathcal{H}_c$ is called the coin-space and $\mathcal{H}_p$ is the position-space.
- If the coin-space is spanned by the basis states $\{|0\rangle, |1\rangle\}$, then we can define the unitary translation of the system $S$ as

$$S = |0\rangle\langle 0| \otimes \sum_i |i+1\rangle\langle i| + |1\rangle\langle 1| \sum_i |i-1\rangle\langle i|$$

- For a walk on the line, $i$ will run over $\mathbb{Z}$. For a walk on a circle, $i$ runs in $\mathbb{Z}_n$

# Balanced Unitary Coin

- If we measure the coin-register after one step, we would wish to obtain a classical unbiased walk, i.e moves to to the right ($|0\rangle$) with probability $\frac{1}{2}$ and to the left ($|1\rangle$) with probability $\frac{1}{2}$
- A frequently used balanced coin is the Hadamard coin $H$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- One iteration of the walk beginning from $|0\rangle |0\rangle$ looks like:

$$\begin{aligned} S(H \otimes I) |0\rangle |0\rangle &= S \left[ \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle \right] \\ &= \frac{1}{\sqrt{2}} (S |0\rangle |0\rangle + S |1\rangle |0\rangle) \\ &= \frac{1}{\sqrt{2}} (|0\rangle |1\rangle + |1\rangle |-1\rangle) \end{aligned}$$

# Difference between Classical and Quantum Walks

- If we continue this process of measuring after every iteration, we would get a classical walk.
- For a walk on the line, the limiting distribution approaches $\mathcal{N}(0, T)$, i.e the expected distance from origin is of the order $\sqrt{T}$
- We wish to let the quantum interference remain.
- For a quantum random walk of $T$ steps, we define the transformation as $U^T$ where

$$U = C \cdot (S \otimes I)$$

- It can be shown that if we start from state $X_0 = |1\rangle |0\rangle$, the classical walk results in $\Pr[X_3 = |-1\rangle] = \frac{3}{8}$ while the quantum walk results in $\Pr[X_3 = |-1\rangle] = \frac{5}{8}$

# Drifting Walks

- The fact that after 3 steps, the probability of being in state $|-1\rangle$ starting from $|1\rangle|0\rangle$ is $\frac{5}{8}$, i.e the walk is drifting to the left.
- This is because the Hadamard coin treats the two basis states differently: it appends a phase of $-1$ only for $|1\rangle$.
- If we started with $|0\rangle|0\rangle$, the walk would drift towards the right.
- To eliminate the asymmetry, start with the state $\frac{1}{\sqrt{2}}(|0\rangle + \iota|1\rangle)$ or use a balanced coin $C$:

$$C = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & \iota \\ \iota & 1 \end{bmatrix}$$

- The quantum walk has variance $\sigma^2 \sim T^2$, i.e the expected distance from origin is of order $T$, quadratically faster!

# Walks on a bounded line

- If there is an absorbing boundary in position $|b\rangle$, we can measure the process at each time step.
- The walk is then given by one step $U$, followed by a measurement w.r.t $|b\rangle$. If we get $|b\rangle$, we stop the walk, else it continues
- The result of the measurement is as follows:

$$M_b |\psi\rangle = \begin{cases} |b\rangle & p_b = |\langle b|\psi\rangle|^2 \\ \frac{|\psi\rangle - \langle b|\psi\rangle |b\rangle}{\sqrt{1-|\langle b|\psi\rangle|^2}} & 1 - p_b \end{cases}$$

- In the classical world, a random walk will always get absorbed and never escape
- In the quantum world, the numerical value of absorption probability changes with the distance to the boundary. The escape probability is non-zero and finite.

# Quantum walks on $d-$regular graphs

- The Hilbert coin-space is now of dimension $d$. Each edge is labelled with $j \in [d]$. Let $e_v^j$ be an edge labelled with index $j$ at vertex $v$. Then the state associated with vertex $v$ pointing along $j$ is given by $|j\rangle |v\rangle$
- The shift is now given by

$$S |j\rangle \otimes |v\rangle = \begin{cases} |j\rangle |w\rangle & \text{if } e_v^j = (v, w) \\ 0 & \text{otherwise} \end{cases}$$

- To obtain a balanced coin, we use the Grover's Diffusion coin (aka unitary DFT coin) given by $G_{ij} = \frac{2}{d} - \delta_{ij}$. Otherwise, the choice of $C$ is guided by the specifics and the symmetries we wish to obtain.
- For a graph which is not regular:
  - Can add self loops
  - Use coins of different dimensions for each vertex

# Reducing a walk using symmetry

- The walk on a hypercube can be reduced to a walk on the line.
- In a $d-$ dimensional hypercube, each vertex is labelled by a $d-$ bit string, and two vertices are connected if their Hamming weight differs by 1.
- If we start from $0^d$, all vertices with equal Hamming weights are interchangeable and at equal distance, and hence, can be cumulated into a single vertex on a line.
- To retain this symmetry, we require a unitary and permutation symmetric coin such as:

$$G_{a,b} = \begin{bmatrix} a & b & b & \dots & b & b \\ b & a & b & \dots & b & b \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ b & b & b & \dots & b & a \end{bmatrix}$$

- This is not a balanced coin! However, it can help us reduce a quantum walk on the hypercube to a quantum walk on the line!

# Walk on the hypercube

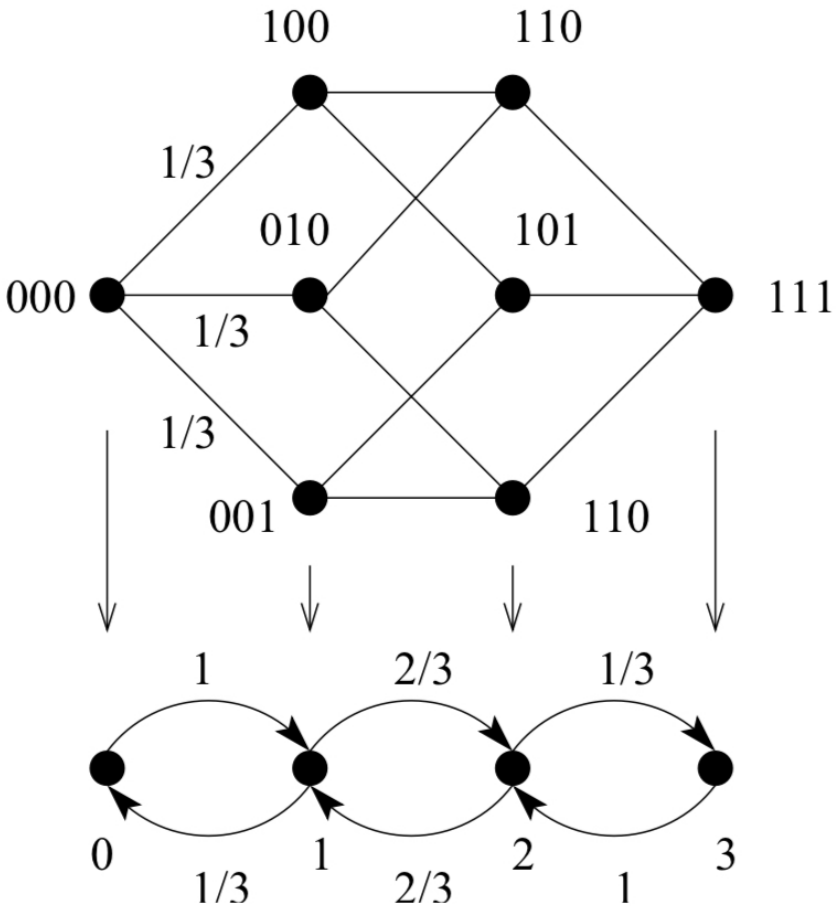



Figure: Reducing the walk on a hypercube using symmetry

# Continous random walks

- We know for a random walk, given a transition matrix $M$

$$\vec{p_{t+1}} = M\vec{p_t}$$

- The transitions transform the states at integer times only.
- Instead, we assume transitions can happen at all times, and the jumping rate is given by $\gamma$, i.e transitions happen between neighbouring states with a probability $\gamma$ per unit time.
- We define the generator matrix as

$$H_{i,j} = \begin{cases} -\gamma & i \neq j \text{ and i and j are connected} \\ 0 & i \neq j \text{ and i and j are not connected} \\ deg(i)\gamma & i = j \end{cases}$$

- Similar to the transition matrix,

$$\frac{dp_i(t)}{dt} = -\sum_j H_{i,j}p_j(t) \implies \vec{p_t} = \exp(-Ht)\vec{p_0}$$

# Another difference between classical and quantum walks

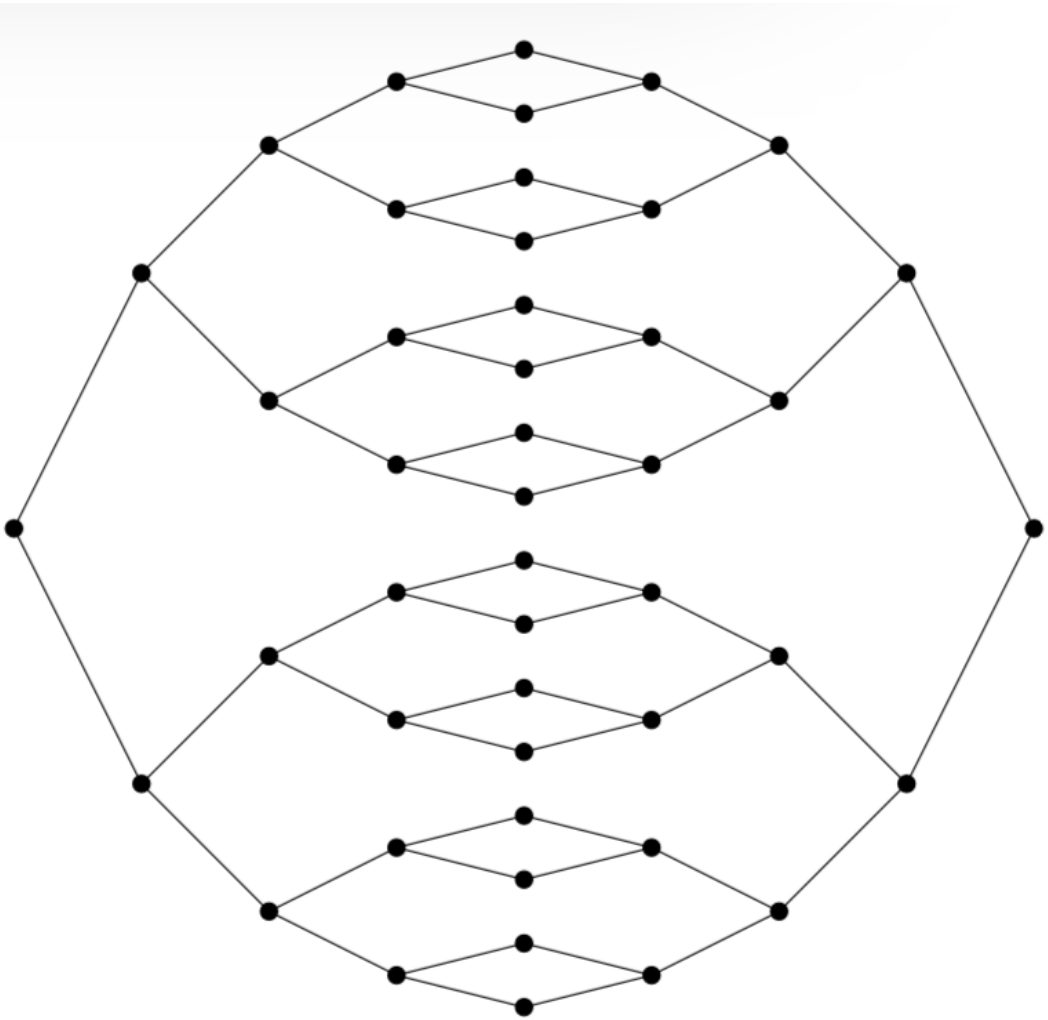


Figure: Graph $G_4$ made by gluing two symmetric binary trees at their leaves.

# Classical walk on $G_4$

- Can reduce the walk to a walk on the line using symmetry.
- There are $2n$ points on the line, with the $i^{th}$ point consisting of $2^i$ vertices if $0 \leq i \leq n$ and $2^{2n-i}$ if $n \leq i \leq 2n$
- Note that if we start from the left node, the probability of moving to column $j+1$ from column $j$ is twice the probability of moving to column $j-1$ if $0 < j < n$
- Similarly, probability of moving to column $j-1$ is twice than that of moving to column $j+1$ if $n < j < 2n$
- If we start at the left root, very easy to reach the middle of the graph, but very difficult to reach the right root! (exponential in n)

# Quantum walk on $G_4$

- $(2n+1)$ dimensional subspace spanned by $|j\rangle$ where $0 \leq j \leq 2n$ as compared to the original $2(1+2+\ldots 2^{n-1})+2^n = 2^{n+1}+2^n-2$ dimensions
- Define $|j\rangle$ to be a superposition over all vertices collapsed into it.

$$|j\rangle = \frac{1}{\sqrt{N_j}} \sum_{a \in col(j)} |a\rangle \text{ where } N_j = \begin{cases} 2^j & 0 \leq j \leq n \\ 2^{n-j} & n \leq j \leq 2n \end{cases}$$

- Recall

$$H_{i,j} = \begin{cases} -\gamma & i \neq j \text{ and i and j are connected} \\ 0 & i \neq j \text{ and i and j are not connected} \\ deg(i)\gamma & i = j \end{cases}$$

# Non-zero elements of Hamiltonian

$$\langle j|\, H\, |j\rangle = \frac{1}{N_j}\left(\sum_{a\in col(j)} H_{aa}^2\right) = \frac{1}{N_j}\sum_{a\in col(j)} deg(a)\gamma$$

$$= \begin{cases} 2\gamma & j = 0, n \text{ or } 2n \\ 3\gamma & o/w \end{cases}$$

$$\langle j|\, H\, |j\pm 1\rangle = \frac{1}{\sqrt{N_j N_{j\pm 1}}}\left(\sum_{a\in col(j), b\in col(j\pm 1)} H_{ab}\right)$$

$$= \frac{1}{\sqrt{N_j N_{j\pm 1}}} - \gamma \times 2N_j$$

$$= -\gamma\sqrt{\frac{N_j}{N_{j\pm 1}}} = -\sqrt{2}\gamma$$

# Quantum walk on $G_4$

- Can be solved using the Bessel function!
- Speed of propogation is linear in time T!
- There exist constants $a, b$ such that for $T \in [an, bn]$, the probability of measuring the walk at a distance $2n + 1$ (i.e right root) is of order $\frac{1}{\sqrt{n}}$

# Results on classical random walks

- Mixing time is given by

$$M_\epsilon = \min\{T | \forall t \geq T \; \forall \vec{p_0} : ||\vec{p_t} - \pi|| < \epsilon\}$$

- Mixing time is related to the gap between largest eigenvalue $\lambda_1$ and second largest $\lambda_2$

$$\frac{\lambda_2}{(1-\lambda_2)\log 2\epsilon} \leq M_\epsilon \leq \frac{1}{1-\lambda_2}(\max_i \log \pi_i^{-1} + \log \epsilon^{-1})$$

- Walk on a $N-$circle:
  - Mixing time is quadratic $M_\epsilon \sim N^2 \log(1/\epsilon)$
  - Expected time to reach vertex $i$ from vertex $j$ with probability 1 is $N^2$
- Walk on $d-$ dimensional hypercube
  - Mixing time $M_\epsilon \sim d \log d \log(1/\epsilon)$
  - Expected time to reach $1^d$ from $0^d$ scales as $2^d$
  - Quantum improvement!

# Results on Quantum Random Walks

- For a classical walk, a stationary distribution is independent of starting state.
- This cannot be true for a quantum random walk since all transformations are unitary, and hence, reversible.
- Instead of studying $\lim_{t\to\infty} \vec{p}^t$, we study the average distribution over all measurement results between 1 and $t$:

$$\vec{c}^t = \frac{1}{t}\sum_{s=1}^{t} \vec{p}^s$$

- $\vec{c}^t$ converges to a stationary distribution.

# Quantum walk on a circle

- Given a state $|\psi\rangle$, the probability of finding it in state $i$ is given by $\vec{p}_i^t = |\langle 0, i|\psi\rangle|^2 + |\langle 1, i|\psi\rangle|^2$
- If the initial state $|\psi_0\rangle = \sum_{k=1}^{2N} a_k |v_k\rangle$ and $\lambda_k, |v_k\rangle$ are the eigenvalues and eigenvectors of $U$, then

$$|\psi_t\rangle = U^t |\psi_0\rangle = \sum_{k=1}^{2N} a_k \lambda_k^t |v_k\rangle$$

- Then,

$$\vec{p}_i^t = \sum_{j=1}^{2N} \sum_{k=1}^{2N} \sum_{\alpha \in \{0,1\}} a_j^* a_k (\lambda_j \lambda_k^*)^t \langle \alpha, i|v_j\rangle \langle v_k|\alpha, i\rangle$$

# Walk on the circle continued

- $$\vec{c_i^t} = \frac{1}{t}\sum_{s=1}^{t}\sum_{j=1}^{2N}\sum_{k=1}^{2N}\sum_{\alpha\in\{0,1\}} a_j^* a_k (\lambda_j \lambda_k^*)^t \langle \alpha, i|v_j\rangle \langle v_k|\alpha, i\rangle$$

- $$\lim_{t\to\infty} \vec{c_i^t} = \sum_{k=1}^{2N}\sum_{\alpha=0}^{1} a_k^* a_k \langle \alpha, i|v_k\rangle \langle v_k|\alpha, i\rangle = \sum_{k=1}^{2N}\sum_{\alpha=0}^{1} |a_k|^2 |\langle \alpha, i|v_k\rangle|^2$$

- This is not independent of initial state because of the presence of coefficients $a_k$.
- The mixing time for classical walks was governed by second largest eigenvector. Here, it is governed by $|\lambda_k - \lambda_l| \forall k, l$
- The mixing time can be bounded by $\frac{N \log N}{\epsilon^3}$, giving quadratic speed up.

# Walk on the hypercube

- The classical walk on hypercube mixeds in time $d \log d$, whereas the quantum walk mixes in $d^{3/2}/\epsilon$ :(
- However, there is one more notion: Hitting time
- In the classical world, hitting times from one corner to the other is exponential
- In the quantum worls, the hitting time from one corner to the other is polynomial in $d$.
- This is an exponential speedup.

# Our attempts - Szegedy's Walk

- Define

$$|\phi_x\rangle = |x\rangle \otimes \sum_{y \in Y} \sqrt{P_{xy}} \, |y\rangle$$

$$|\psi_y\rangle = \sum_{x \in Y} \sqrt{P_{yx}} \, |x\rangle \otimes |y\rangle$$

- Then one step of the walk is defined as

$$W = R_2 R_1$$

$$R_2 = 2 \sum_{y \in Y} |\psi_y\rangle\langle\psi_y| - I$$

$$R_1 = 2 \sum_{x \in Y} |\phi_x\rangle\langle\phi_x| - I$$

You would recognize this as the reflection operators across the states $|\phi_x\rangle$ and $|\psi_y\rangle$

# Our attempts- Szegedy's walk on a weighted graph

- For a weighted graph, we simply normalzize all the weights going out of a vertex.
- This results in the following superposition at a particular vertex:

$$|\phi_v\rangle = |v\rangle \otimes \sum_{u \in N(v))} \sqrt{P_{vu}}\,|u\rangle = |v\rangle \otimes \frac{1}{\sqrt{\sum_{t \in N(v)} w_{vt}}} \sum_{u \in N(v))} \sqrt{w_{vu}}\,|u\rangle$$

- It can be shown that one step of Szegedy's walk is equivalent to two steps of a coined walk. In fact, $R_1$ is equivalent to $C$ and $R_2$ is equivalent to $SCS$.

# Our attempts- Szegedy's walk on a weighted graph

- If the state of a system is $|\psi\rangle$ where

$$|\psi\rangle = \sum_v \otimes \sum_{u \in N(v)} \alpha_{vu} |u\rangle$$

then

$$C|\psi\rangle = \sum_v \otimes \sum_{u \in N(v)} (2\bar{\alpha}_v \sqrt{w_{vu}} - \alpha_{vu}) |u\rangle$$

where

$$\bar{\alpha}_v = \frac{\sum_{u \in N(v)} \sqrt{w_{vu}} \alpha_{vu}}{\sum_{u \in N(v)} w_{vu}}$$

- In one sense, inversion about weighted mean!

# Coined walks on Directed graphs

- If you can walk from $a$ to $b$, there has to exist a path back from $b$ to $a$.
- Partition the graphs into cycles such that each edge is part of atleast one cycle.
- Perform a walk on the space where each basis state represents a cycle.
- In case the graph is not "reversible", i.e there exist $a$ and $b$ such that there is a path from $a$ to $b$ but not from $b$ to $a$, partition the graphs into "reversible subgraphs", perform walks inside each subgraph, and transition from one subgraph to another using irreversible edges.

# Quantum walks on Directed and Weighted Graphs

- Considering both the walks equivalent, we tried to evolve the states using both notions of the walk, and equated the probabilities of being in a state after a step.
- Math worked out well, conceptual issues arose!

# Simplistic model checking using quantum walks

- The combination of the ideas were to be used simple model checking such as checking an until condition.
- Came up with naive solutions which would require quantum memory and backtracking.
- Turns out lots of work already done on this, with concepts and ideas well beyond us!

# References


- J. Kempe, "Quantum random walks: An introductory overview," *Contemporary Physics*, vol. 44, p. 307–327, July 2003.
- A. Montanaro, "Quantum walks on directed graphs," 2005.
- T. G. Wong, "Equivalence of szegedy's and coined quantum walks," *Quantum Information Processing*, vol. 16, July 2017.
- S. A. Ortega and M. A. Martin-Delgado, "Discrete-time semiclassical szegedy quantum walks," *Physica A: Statistical Mechanics and its Applications*, vol. 625, p. 129021, Sept. 2023.
- M. Szegedy, "Quantum speed-up of markov chain based algorithms," in *45th Annual IEEE Symposium on Foundations of Computer Science*, pp. 32–41, 2004.